

Total No. of Questions :10]

SEAT No. :

P3222

[5461]-263

[Total No. of Pages : 2

B.E. (Computer Engineering)
SMART SYSTEM DESIGN & APPLICATIONS
(2012 Pattern) (Semester-I) (End Sem.) (410443)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) *Neat diagram must be drawn wherever necessary.*
- 2) *Figures to the right indicate full marks.*
- 3) *Assume suitable data if necessary.*

- Q1)** a) Explain the A* search algorithm with the help of suitable example. **[8]**
b) What is alpha beta pruning? Explain alpha beta search algorithm with suitable example. **[6]**
c) Explain the architecture of Goal Based agents. **[4]**

OR

- Q2)** a) Explain the environment types & PEAS properties of agent? **[8]**
b) Explain game theory and knowledge structure? **[6]**
c) Explain the unification algorithm and state its application. **[4]**

- Q3)** a) What is baye's rule? Write down its application. **[4]**
b) Explain the concept of uncertainty? Write down an example illustrating the behavior of an agent in an uncertain world. **[6]**
c) Compare and contrast propositional logic and FOL. **[4]**

OR

- Q4)** a) Explain Role of NLP in AL? **[4]**
b) What is Expert System shell? Why explanation is necessary in expert system? **[6]**
c) Explain support vector Machine with issues and applications. **[4]**

P.T.O.

- Q5)** a) What are the basic axioms of probability? Explain how to derive the useful facts from the basics axioms with suitable example. [6]
b) Write a short note on: [8]
i) Information Retrieval
ii) Information Extraction

OR

- Q6)** a) How to represent and Evaluate decision problem with a decision network. [8]
b) What is prior probability and posterior probability? Explain with suitable example. [6]
- Q7)** a) What is supervised learning? Explain any one. [6]
b) What is Artificial Neural Network? Explain its types. [6]

OR

- Q8)** a) Write a short note on Wumpus world environment. [6]
b) Write a note on Robotics software architecture. [6]
- Q9)** a) Explain in details the components that help in reconstructing the world in 3D. [6]
b) Write a short note on: [6]
i) Dynamic Bayesian Network
ii) Kalman Filters

OR

- Q10)** a) List application domains of robotics. Explain any one in detail. [6]
b) How to represent and evaluate decision problem with a decision network. [6]

